

**SMART INDIA
HACKATHON
2026**

- **Problem Statement Title - Use case on web-based integrated project-monitoring platform**
- **Theme - Smart Automation**
- **PS Category - Software**
- **Team ID - [To be provided]**
- **Registered Team Name - [To be provided]**

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PARAKH: Evidence-led early warning

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

CURRENT PROJECT EXPLORER • JULY 2026

PARAKH turns monthly PAIMANA Flash Reports into a searchable project portfolio. A transparent rule score describes current signals; experimental models estimate cost or schedule changes by the next report.

WHO USES IT

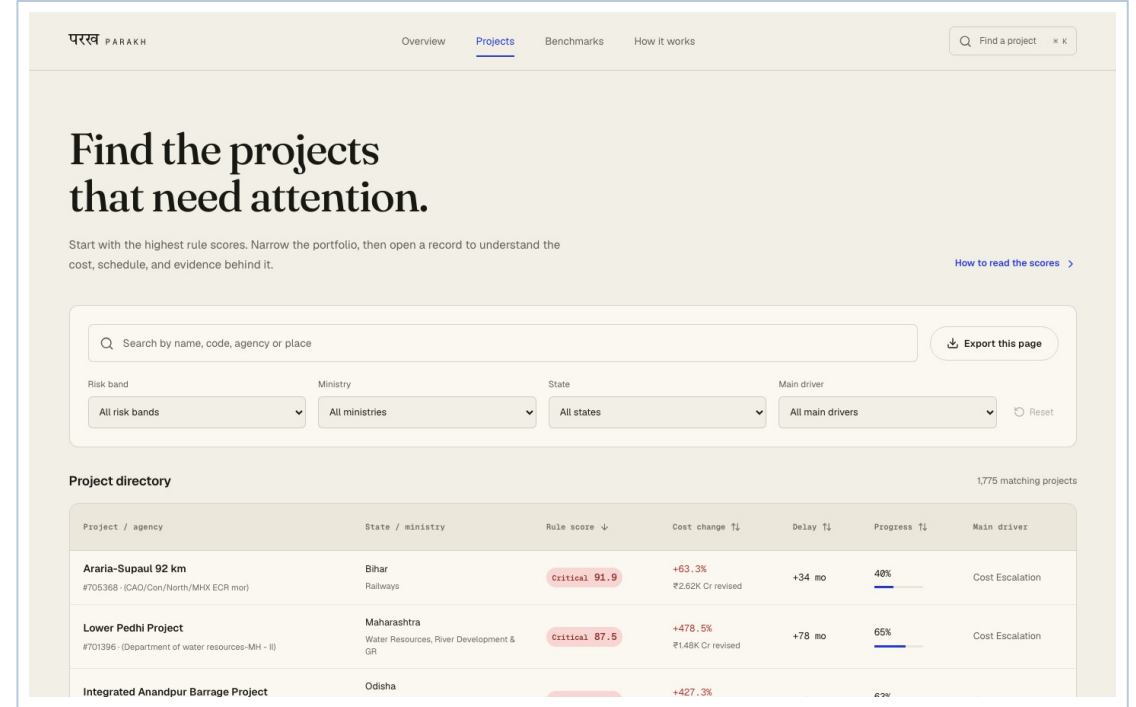
MoSPI/IPMD analysts • line ministry reviewers • project administrators

PROTOTYPE TODAY

Four reports (Apr–Jul 2026); the July snapshot contains 1,775 projects.

NEXT BUILD

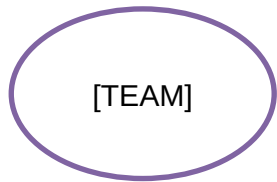
Automatic quality gate, source-page evidence, and a human review queue.



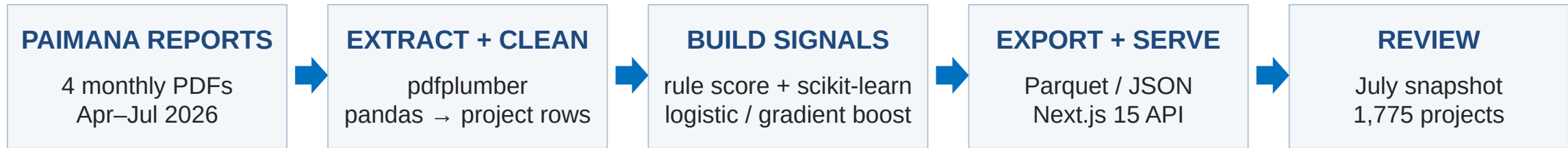
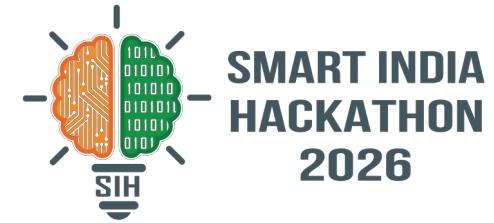
The screenshot shows the PARAKH Project Explorer interface. At the top, there's a navigation bar with 'परख PARAKH', 'Overview', 'Projects' (active), 'Benchmarks', and 'How it works'. A search bar on the right says 'Find a project'. Below the navigation bar, a heading reads 'Find the projects that need attention.' followed by a subtext: 'Start with the highest rule scores. Narrow the portfolio, then open a record to understand the cost, schedule, and evidence behind it.' and a link 'How to read the scores >'. A search bar with the placeholder 'Search by name, code, agency or place' and an 'Export this page' button is present. Below the search bar are four filter dropdowns: 'Risk band' (All risk bands), 'Ministry' (All ministries), 'State' (All states), and 'Main driver' (All main drivers), with a 'Reset' button. The 'Project directory' section shows '1,775 matching projects'. The table below lists projects with columns: Project / agency, State / ministry, Rule score ↓, Cost change ↑↓, Delay ↑↓, Progress ↑↓, and Main driver.

Project / agency	State / ministry	Rule score ↓	Cost change ↑↓	Delay ↑↓	Progress ↑↓	Main driver
Araria-Supaul 92 km #705368 - (CAO/Con/North/MHX ECR mor)	Bihar Railways	Critical 91.9	+63.3% ₹2.62K Cr revised	+34 mo	40%	Cost Escalation
Lower Pedhi Project #701396 - (Department of water resources-MH - II)	Maharashtra Water Resources, River Development & GR	Critical 87.5	+478.5% ₹1.68K Cr revised	+78 mo	65%	Cost Escalation
Integrated Anandpur Barrage Project	Odisha		+427.3%			

Working screen: search, filters, project measures and rule score. Model estimates remain experimental.



TECHNICAL APPROACH



Open-source stack: Python • pandas / NumPy • scikit-learn • Parquet • Next.js 15 / React 19 • Recharts

April audit: 1,981 project rows and all three portfolio totals reconcile to the MoSPI report (0.0% displayed difference). checker.py is separate today; making it block exports is next.

ONE TEMPORAL HOLDOUT • JUNE INPUT → JULY OUTCOMES • NEXT MONTH

SCHEDULE-DATE CHANGE

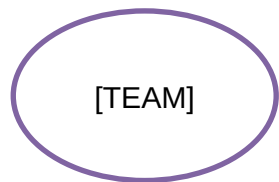
Gradient boosting ROC-AUC 0.767 • PR-AUC 0.342
Logistic regression ROC-AUC 0.728 • PR-AUC 0.280
1,732 labelled rows • 268 schedule changes

COST REVISION

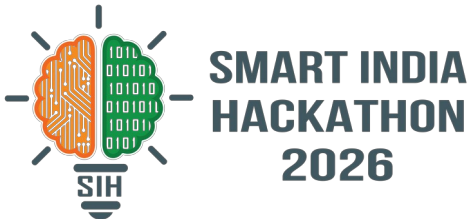
Gradient boosting ROC-AUC 0.436 • PR-AUC 0.039
Logistic regression ROC-AUC 0.503 • PR-AUC 0.044
1,732 labelled rows • 72 cost revisions

Cost model: not decision-ready

One temporal test month; probabilities are uncalibrated. These results are preliminary, not deployment validation.



FEASIBILITY AND VIABILITY



TECHNICAL

Open-source Python + Next.js already used. Batch report processing; no GPU or paid model API is configured.

ECONOMIC

No new database or API subscription for this prototype. Hosting, support and review costs still need an operational estimate.

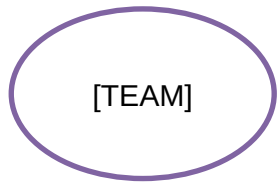
OPERATIONAL

Fits monthly report review. Analysts retain authority; there is no ministry alert or intervention integration.

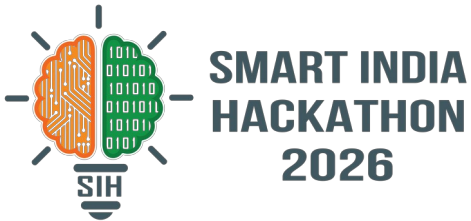
CHALLENGE → RESPONSE

Only four report months	Add older reports; repeat time-forward validation across multiple months.
PDF layouts can shift	Add per-page schema and total checks; wire checker.py as a stop/go gate before export.
Cost model is weak	Keep it out of decisions until it beats a baseline repeatedly and probabilities are calibrated.
Warnings need context	Show source month and field evidence; an analyst confirms or dismisses before follow-up.

Pilot acceptance: reconciliation coverage • top-K warning precision • lead time • reviewer effort



IMPACT AND BENEFITS



OFFICIAL PAIMANA PORTFOLIO • APRIL 2026

1,981

projects • 17 ministries / departments

₹42.78 lakh cr

revised project cost

₹20.36 lakh cr

spent • 47.59% of revised cost

WHAT CHANGES FOR REVIEWERS

TODAY • The prototype turns the July 2026 snapshot (1,775 projects) into a searchable, filterable directory with a transparent rule score and project details.

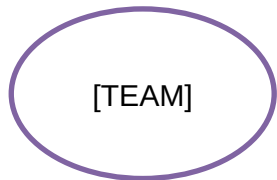
INTENDED • Surface candidates for earlier review and show why. Time saved, earlier detection and fewer delays have not been measured.

ANALOGY

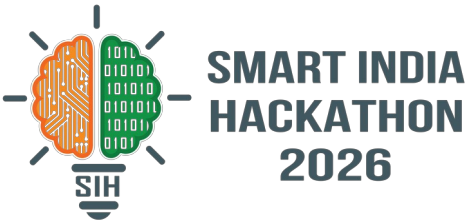
Like an inspection triage board: it points to the project and its evidence; the authorized official decides what to inspect next.

Potential value: prioritized follow-up • traceable reasoning • open-source reuse

No savings, delay reduction or avoided environmental impact has been measured.



RESEARCH AND REFERENCES



Evidence used in this proposal

[MoSPI • PAIMANA Flash Report • April 2026](#)

Pages 1–3 and Table 6: 1,981 projects; ₹37.13 lakh crore original cost; ₹42.78 lakh crore revised cost; ₹20.36 lakh crore expenditure.

[MoSPI • PAIMANA project-monitoring portal](#)

Official report page and public project dashboard used to understand report fields and the monthly reporting context.

[SIH 2026 • Problem Statement 26103 • MoSPI / DIID](#)

Official challenge scope supplied by the team: predictive analytics, model comparison, CUF-field assessment and decision support.

[PARAKH prototype • project source and evaluation artifacts](#)

Four monthly source PDFs (April–July 2026); src/checker.py; src/model_train.py; data/processed/model_metrics.json.

The prototype uses four report months only. The current code repository link is configured from the team's Git remote; confirm public access before submission.